PREDICTIVE ANALYTICS

MINOR PROJECT REPORT

(FIFTH SEMESTER)

MENTALBRIDGE

Student Stress Analysis and Prediction Using Predictive Analytics

Submitted by

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Section: QR

Course Code: INT 234

Under the Guidance of

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DISCIPLINE OF CSE/IT

LOVELY PROFESSIONAL UNIVERSITY, PHAGWARA

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DECLARATION

I, Arya Srivastava, student of B.Tech CSE (Data Science), Section QR, Registration Number 12406211, under the CSE/IT Discipline at Lovely Professional University, Punjab, declare that this report titled "MentalBridge: Student Stress Analysis and Prediction Using Predictive Analytics" has been prepared as part of the Predictive Analytics minor project for Course Code INT 234.

The report describes the dataset, exploratory analysis, preprocessing steps, machine-learning models and observations made during the project. The analysis and reported numerical results are based on the supplied student stress dataset and the experiments documented in the accompanying Jupyter Notebook.

I understand that the project is an academic exercise in data analysis and machine learning. Its predictions are not a medical diagnosis and should not be used as a substitute for professional advice. I have made efforts to describe the methods and limitations in a clear manner.

Date: 31 October 2026

Place: Phagwara, Punjab

Signature: _____

Name: Arya Srivastava

CERTIFICATE

This is to certify that Arya Srivastava, bearing Registration Number 12406211, student of B.Tech CSE (Data Science), Section QR, has completed the Predictive Analytics minor project titled “MentalBridge: Student Stress Analysis and Prediction Using Predictive Analytics” for Course Code INT 234 under the guidance and supervision of Dr. Mrinalini Rana.

To the best of my knowledge, the project work presented in this report is based on the student's study and effort carried out for the academic requirements of the course.

Signature and Name of the Supervisor: _____

Dr. Mrinalini Rana

Designation: Faculty

U.ID: 22138

School of Technology, Lovely Professional University

Phagwara, Punjab

Date: ___31-10-2026_____

Note: Supervisor signature and certification date are intentionally left for the faculty member.

ACKNOWLEDGEMENT

I would like to express my sincere thanks to my project supervisor, Dr. Mrinalini Rana, for guiding me during this Predictive Analytics minor project. Her academic guidance helped me understand how the concepts discussed in class can be applied to a real dataset.

I am thankful to the faculty members of the CSE/IT discipline and Lovely Professional University for providing the learning environment and resources required to complete this work. The practical exercises shared during the course were useful while implementing data loading, preprocessing, visualisation and classification algorithms.

I would also like to acknowledge the creators of the Stress Indicators Dataset for Mental Health Classification hosted on Mendeley Data. The dataset provided the survey records used for the analysis in this report.

Finally, I thank my classmates, friends and family for their support and encouragement throughout the project. Preparing this report helped me practise Python libraries and understand the importance of checking data quality before training a model.

Arya Srivastava

ABSTRACT

MentalBridge is a Predictive Analytics project that examines stress-related indicators in a survey dataset and compares four classification algorithms. The project uses Python libraries including Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn. The workflow includes data loading, data-quality checks, exploratory data analysis, feature selection, an 80:20 train-test split and classification model evaluation.

The notebook loads 2,000 records with 26 columns. It identifies 1,191 exact duplicate rows but does not remove them; all 2,000 records remain in the analysis and model training. No missing values are found. The target variable is stress_type, with coded classes 0, 1 and 2. The feature stress_experience is excluded from the predictors, leaving 24 input features.

Four classifiers are trained: Support Vector Machine (RBF kernel), Gaussian Naive Bayes, K-Nearest Neighbors (k=5) and Random Forest (100 trees). The data is split into 1,600 training records and 400 testing records using test_size=0.20 and random_state=42, without stratification. Test accuracies are 99.00% for SVM, 94.50% for Naive Bayes, 94.75% for KNN and 99.50% for Random Forest. Random Forest achieves the highest accuracy in this run.

The target is highly imbalanced: class 1 accounts for 1,828 of the 2,000 records. Duplicate records are retained in the notebook workflow, so the test set may contain records identical to training records. The reported accuracy should therefore be interpreted cautiously and is not evidence of equivalent performance on independent new students. This project is an academic exercise, not a medical or psychological diagnosis.

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All numerical results in this report are taken from the supplied Jupyter Notebook and its accompanying output PDF. The notebook retains all 2,000 rows after detecting duplicates.

1. INTRODUCTION

Students manage several responsibilities during their academic life. Classes, assignments, examinations, peer interactions, personal responsibilities and changes in routine can influence how students feel and function. Stress-related survey data can be studied to understand how different indicators appear together in a group of responses. This project uses Predictive Analytics methods to explore such patterns.

MentalBridge focuses on a survey dataset in which each record contains demographic information and ratings for different physical, psychological, academic and social indicators. The target column, stress_type, is a coded category. The project studies the available variables and trains classification models to estimate the target category from the other selected features.

The work is organised in a practical sequence. First, the CSV file is loaded with Pandas and inspected. Missing values and exact duplicate rows are checked. The notebook reports zero missing values and 1,191 duplicate rows; it does not drop those duplicates. Visualisations are then used to examine distributions and relationships. Finally, four classification algorithms are trained and compared using a separate test subset.

The project name MentalBridge represents the link between data analysis and student well-being awareness. It does not mean that the model can assess a person's mental health condition. The dataset and models are used only for academic exploration.

1.1 BACKGROUND OF THE STUDY

Predictive Analytics uses past observations to identify patterns and estimate an outcome. In a classification problem, the outcome belongs to a set of categories. Here, stress_type is categorical, so classification is more suitable than ordinary linear regression. The input variables are survey indicators, and the model learns associations between these inputs and the coded target classes.

Exploratory Data Analysis is important before modelling. A dataset may contain repeated rows, missing values, unusual values or unequal class counts. If these issues are ignored, a model can appear more accurate than it would be on genuinely new records. For this reason, the project checks duplicates and class distribution before training.

The project also uses visual analysis. Histograms show the distribution of individual variables. Boxplots show spread and possible outliers. Scatterplots help compare two variables, while a correlation heatmap summarises pairwise linear associations between numerical columns. These charts do not prove cause and effect, but they help guide interpretation.

The four selected classifiers represent different approaches: SVM searches for separating boundaries, Naive Bayes uses probability, KNN compares nearby observations, and Random Forest combines multiple decision trees.

1.2 PROBLEM STATEMENT

The main problem addressed in this project is how to examine student stress-related survey indicators and compare basic machine-learning classifiers for predicting the coded stress_type category. The dataset contains several types of indicators, so a structured analysis is required before model development.

A second issue is data quality. The duplicate check identifies 1,191 exact duplicate rows. However, the notebook does not call drop_duplicates(), so these rows remain in the dataset used for EDA and modelling. Consequently, identical records may appear in both training and testing subsets after the random split, which can inflate the apparent test performance. This is an important limitation of the current experiment.

A third issue is class imbalance. The target counts are class 0: 70, class 1: 1,828 and class 2: 102. Class 1 represents the large majority of the dataset. Accuracy is reported because it is the metric used in the notebook to select the best-fit model, but the confusion matrices should also be considered when interpreting the minority classes.

The project does not attempt to diagnose stress or provide clinical advice. It is limited to patterns and predictions within the supplied dataset.

1.3 OBJECTIVES OF THE PROJECT

The project has five objectives. These objectives were selected to connect dataset exploration with the Predictive Analytics topics covered in the course.

23. Demographic analysis: examine the age distribution and gender-category counts in the survey data.
24. Physical indicator analysis: explore variables such as heartbeat palpitations, sleep problems, headaches and weight changes.
25. Psychological indicator analysis: study variables such as anxiety/tension, restlessness, sadness/low mood, irritability and concentration problems.
26. Relationship analysis: examine how selected variables, including anxiety and sleep problems and academic overload and stress_type, vary together.
27. Classification and model comparison: train SVM, Naive Bayes, KNN and Random Forest, compare their test accuracy and identify the highest-accuracy model.

The objectives are descriptive and predictive. They do not claim that any one survey indicator causes stress. The results describe the records available in the dataset.

1.4 SCOPE AND LIMITATIONS

The scope of MentalBridge is limited to the supplied survey dataset and the four classification algorithms implemented in the notebook. The analysis includes descriptive statistics, distributions, skewness, correlations and selected relationships. The classification task uses 24 input features after excluding stress_type as the target and stress_experience as a potential leakage variable.

The dataset contains coded values. The available file does not provide a verified codebook in the uploaded materials that explains every numeric category in detail. Therefore, the report refers to values as recorded scores or coded categories and avoids assigning unsupported meanings to gender codes or stress_type classes.

The dataset is imbalanced and contains 1,191 exact duplicate rows. Although the duplicates are detected, they are not removed in the notebook. The analysis and model evaluation therefore use all 2,000 rows. The results are based on one non-stratified random 80:20 train-test split with random_state=42. A different split, deduplicated data or an independent external dataset may produce different results.

The study is not a clinical assessment, a screening tool or a medical recommendation system. No period-management or menstrual-cycle prediction is performed because the supplied dataset does not contain the required cycle dates or related observations.

2. SOURCE OF DATASET

The project uses the dataset titled "Stress Indicators Dataset for Mental Health Classification", hosted on Mendeley Data. The dataset was supplied as a CSV file and loaded directly into the notebook using the Pandas read_csv() function.

Dataset source: Mendeley Data. Dataset page: https://data.mendeley.com/datasets/2gsjv8m7ch/2. The source page should be consulted for the dataset description, version information, creator details and licence conditions when the report is submitted.

The CSV used in this analysis contains 2,000 rows and 26 columns. The columns include demographic information, stress-related experience, physical and psychological indicators, academic and social indicators, and the target column stress_type.

The dataset is treated as a secondary dataset for academic analysis. No new survey responses were collected by the student for this experiment. The notebook detects 1,191 exact duplicate rows but retains them; therefore, the 2,000 records should not be assumed to represent 2,000 independent participants.

Dataset item	Value
Original file rows	2,000
Columns	26
Exact duplicate rows detected	1,191
Rows retained for analysis	2,000
Missing values	0
Target column	stress_type
Input features used	24

2.1 DATASET DESCRIPTION AND VARIABLES

The dataset includes 26 columns. One column, stress_type, is used as the prediction target. The column stress_experience is excluded from model inputs because it may be closely connected to the target and could introduce target leakage. The remaining 24 columns are used as predictors.

Variable group	**Examples from the dataset**
Demographic	gender, age
Physical indicators	heartbeat_palpitations, sleep_problems, headaches, health_issues, weight_changes
Psychological indicators	anxiety_tension, restlessness, irritability, concentration_problems, sadness_low_mood, loneliness_isolation
Academic and social indicators	academic_overload, peer_competition, relationship_stress, professor_difficulties, work_environment, home_environment
Target	stress_type

Most indicator columns contain small integer-coded values. Age is numeric and has a wider range than the rating variables. The report uses the original column names in code and gives readable descriptions in the text. This makes it easier to connect each explanation with the corresponding notebook cell.

2.2 OVERALL METHODOLOGY

The project follows a predictive analytics workflow. The dataset is loaded and inspected. Data quality is checked through missing-value counts and duplicate detection. The notebook finds no missing values and 1,191 exact duplicate rows; it does not remove the duplicates. EDA is performed on all 2,000 records to examine the target distribution, demographics, selected indicator distributions and relationships between variables.

After EDA, the target and input features are separated. The target is stress_type. The stress_experience column is excluded from X. The dataset is divided into training and testing subsets using test_size=0.20 and random_state=42. No stratify argument is supplied, so the split is not stratified. This produces 1,600 training records and 400 test records.

StandardScaler is fitted on the training data for the models that use scaled inputs. The fitted scaler is then applied to the test data. Random Forest is trained on the original numeric feature values because tree-based models do not require standard scaling in the same way as distance- and margin-based models.

The four models generate predictions for the same 400-record test set. The notebook compares the classifiers using accuracy and selects the model with the highest accuracy. It does not calculate precision, recall, F1-score, log loss or ROC-AUC in the supplied implementation.

3. DATA QUALITY AND CLEANING

The original dataset contains 2,000 records. The missing-value check returned zero missing values across all 26 columns. The duplicate check identified 1,191 exact duplicate rows. The notebook uses df.duplicated().sum() to report this count, but does not apply drop_duplicates().

All 2,000 records and 26 columns remain in the DataFrame after the checks. No rows are removed during the notebook's data-quality stage. This means duplicate records may be present in both the training and testing subsets after the random split.

The original target counts are class 0: 70, class 1: 1,828 and class 2: 102. These counts sum to 2,000 and show a substantial class imbalance, with class 1 forming the majority. The report therefore interprets accuracy alongside the class-wise counts shown in the confusion matrices.

Cleaning check	Finding	Action
Missing values	0 across all columns	No imputation needed for current data
Exact duplicate rows	1,191 detected	Retained; drop_duplicates() is not called
Target column	stress_type	Retained as y
Potential leakage column	stress_experience	Excluded from X

3.1 DESCRIPITIVE ANALYSIS

```
df = pd.read_csv("/Users/aryasrivastava23/Documents/Stress Indicators Dataset for Mental Health Classification.csv")

print("Shape of dataset:", df.shape)

df.head(10)
```

Python

```
Shape of dataset: (2000, 26)
```

	gender	age	stress_experience	heartbeat_palpitations	anxiety_tension	sleep_problems	restlessness	headaches	irritability	concentration_problems	...
0	1	20	1	2	2	4	3	1	4	5	...
1	0	19	4	5	5	4	4	5	4	5	...
2	0	21	4	2	3	3	4	4	1	3	...
3	0	20	1	3	2	1	3	1	5	5	...
4	0	19	4	3	3	2	3	4	2	3	...
5	0	19	2	3	2	3	4	1	1	1	...
6	0	18	1	1	1	4	3	3	1	3	...
7	0	19	1	1	4	1	2	3	2	3	...
8	0	21	3	3	4	3	2	2	3	5	...
9	0	19	3	2	3	2	3	3	4	5	...

10 rows × 26 columns

	gender	age	stress_experience	heartbeat_palpitations	anxiety_tension	sleep_problems	restlessness	headaches	irritability	concentration_problems	...	professor_difficulties
count	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	2000.000000	...	2000.000000
mean	0.354000	20.007500	2.975000	2.739500	2.532000	2.767000	2.660500	2.617000	2.701500	2.695000	...	2.453000
std	0.478329	4.776385	1.127391	1.104649	1.201956	1.257578	1.284159	1.256225	1.321844	1.314473	...	1.196874
min	0.000000	14.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	1.000000	...	1.000000
25%	0.000000	19.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	2.000000	...	1.000000
50%	0.000000	19.000000	3.000000	3.000000	2.000000	3.000000	2.000000	2.000000	3.000000	3.000000	...	2.000000
75%	1.000000	20.000000	4.000000	4.000000	3.000000	4.000000	4.000000	3.250000	4.000000	4.000000	...	3.000000
max	1.000000	100.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	5.000000	...	5.000000

rows x 26 columns

3.2 EDA: DEMOGRAPHIC ANALYSIS

The demographic analysis focuses on age and gender using all 2,000 records. The age histogram shows the distribution of numeric age values, while the gender count plot displays the number of records in each coded gender category.

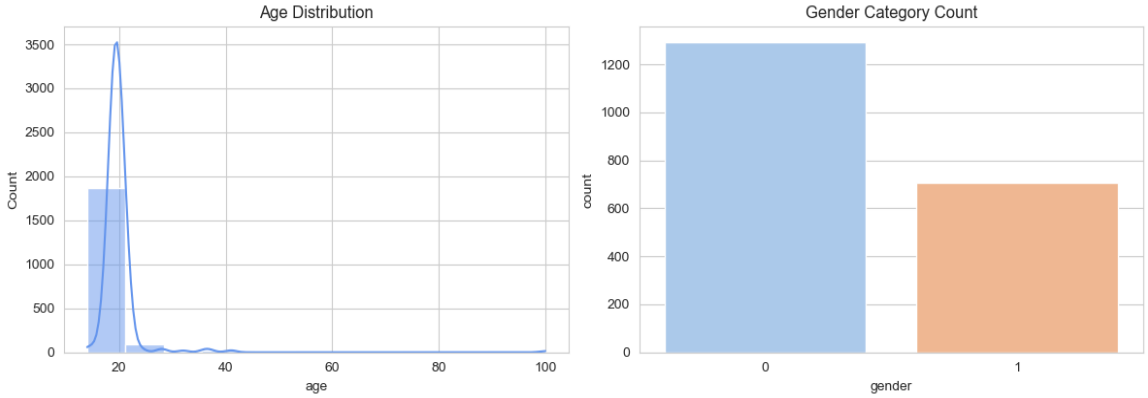


Figure 2. Age distribution and gender category count for all 2,000 records.

The gender variable is displayed using its numeric codes because a confirmed codebook was not available in the materials supplied for this report. The chart should therefore be read as a count of coded categories rather than as a labelled male/female comparison.

The age distribution is strongly right-skewed: the notebook reports a maximum age of 100 and skewness of approximately 12.49. This extreme value stretches the horizontal axis and compresses the main concentration of ages. The chart is descriptive and does not establish that the dataset represents all university students.

3.3 EDA: DISTRIBUTION AND SKEWNESS

Histograms were created for anxiety_tension, sleep_problems, restlessness and academic_overload using all 2,000 records. These variables use a limited range of integer scores, so the bars show how frequently each score occurs and whether responses are concentrated around lower, middle or higher values.

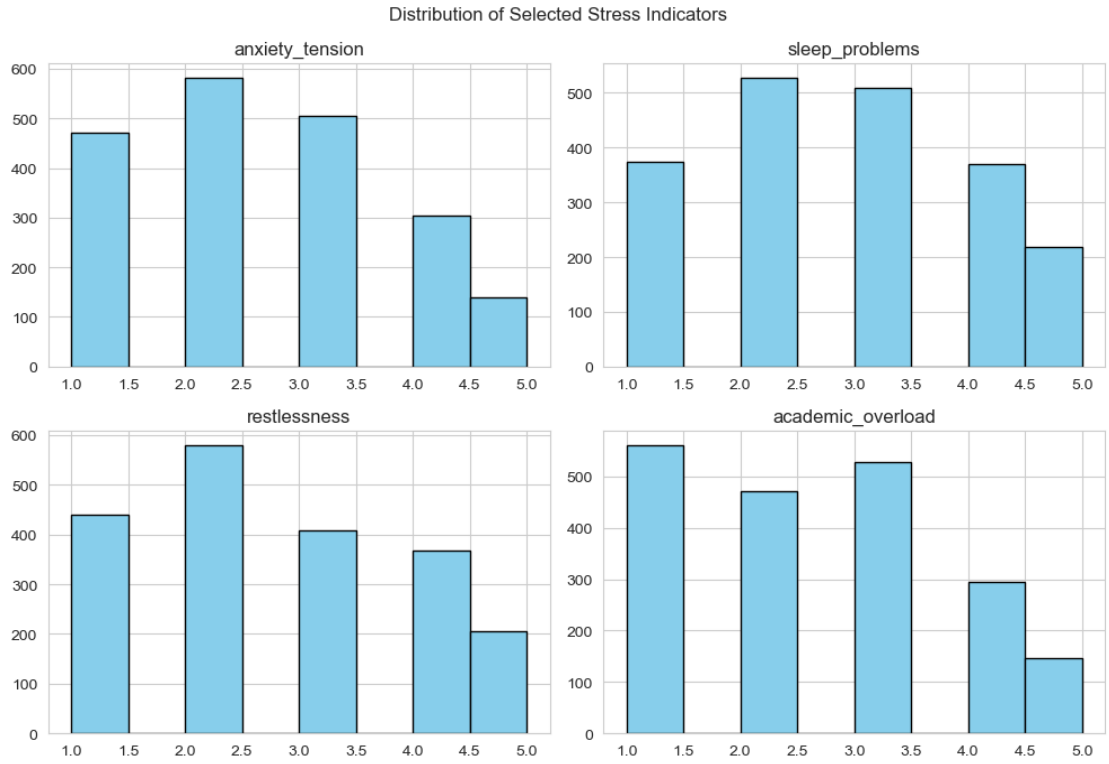
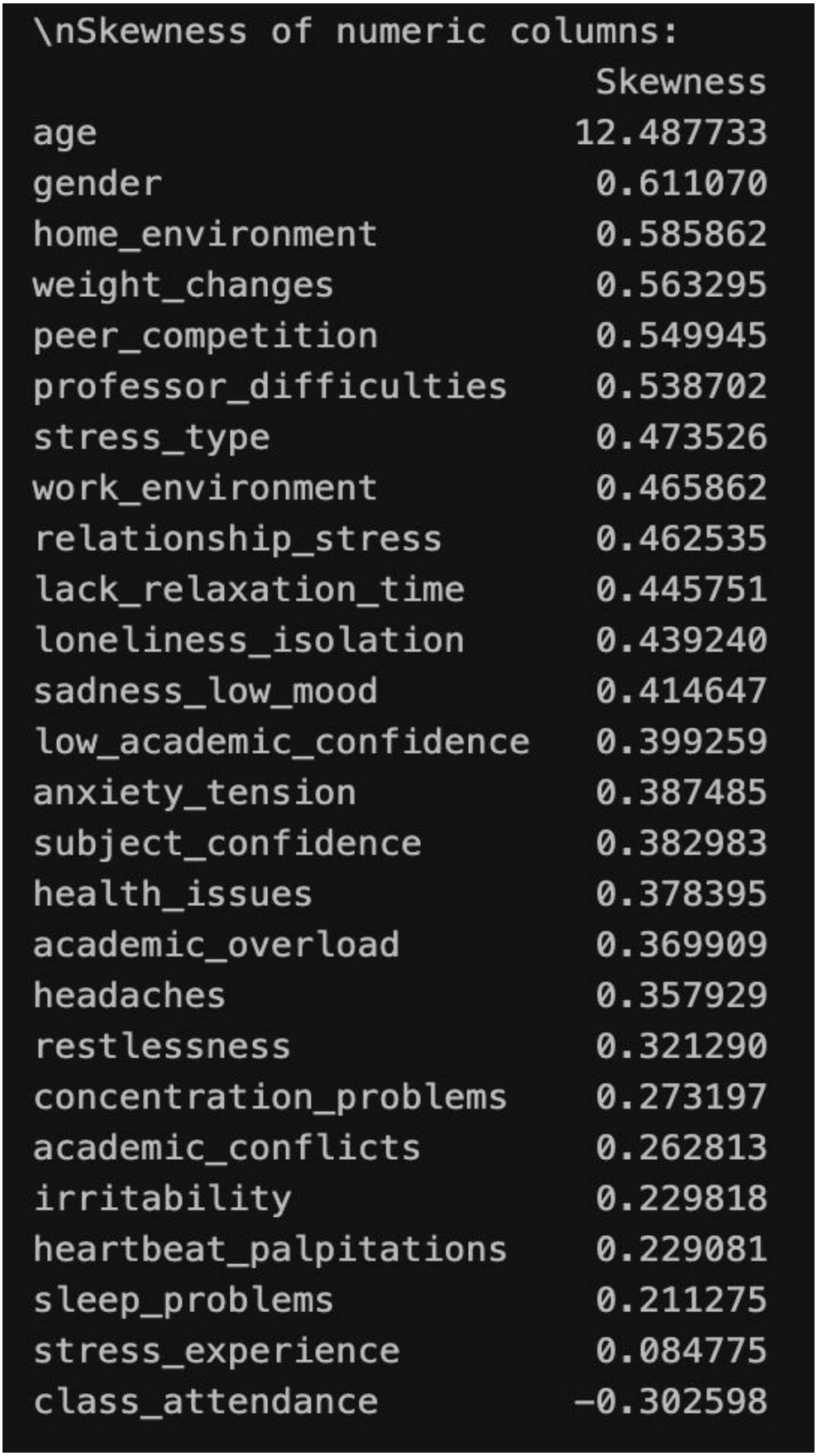


Figure 3. Histograms of selected stress indicators.

Skewness was calculated for all numeric columns. Age has the largest positive skewness (12.487733), while class_attendance has negative skewness (-0.302598). Most other variables have comparatively modest positive skewness. Skewness describes distribution asymmetry and does not by itself indicate whether a variable is useful for prediction.



```
\nSkewness of numeric columns:
                            Skewness
age                        12.487733
gender                      0.611070
home_environment            0.585862
weight_changes              0.563295
peer_competition            0.549945
professor_difficulties      0.538702
stress_type                 0.473526
work_environment            0.465862
relationship_stress         0.462535
lack_relaxation_time        0.445751
loneliness_isolation        0.439240
sadness_low_mood            0.414647
low_academic_confidence     0.399259
anxiety_tension             0.387485
subject_confidence          0.382983
health_issues               0.378395
academic_overload           0.369909
headaches                   0.357929
restlessness                0.321290
concentration_problems      0.273197
academic_conflicts          0.262813
irritability                0.229818
heartbeat_palpitations      0.229081
sleep_problems              0.211275
stress_experience           0.084775
class_attendance           -0.302598
```

Figure 4. Skewness of numeric columns in the original dataset.

3.4 EDA: SPREAD OF PHYSICAL AND PSYCHOLOGICAL INDICATORS

A boxplot was used to compare the spread of heartbeat_palpitations, sleep_problems, headaches, anxiety_tension and sadness_low_mood across all 2,000 records. The median is shown by the line inside each box. The box represents the interquartile range, and points beyond the whiskers may be displayed as potential outliers.

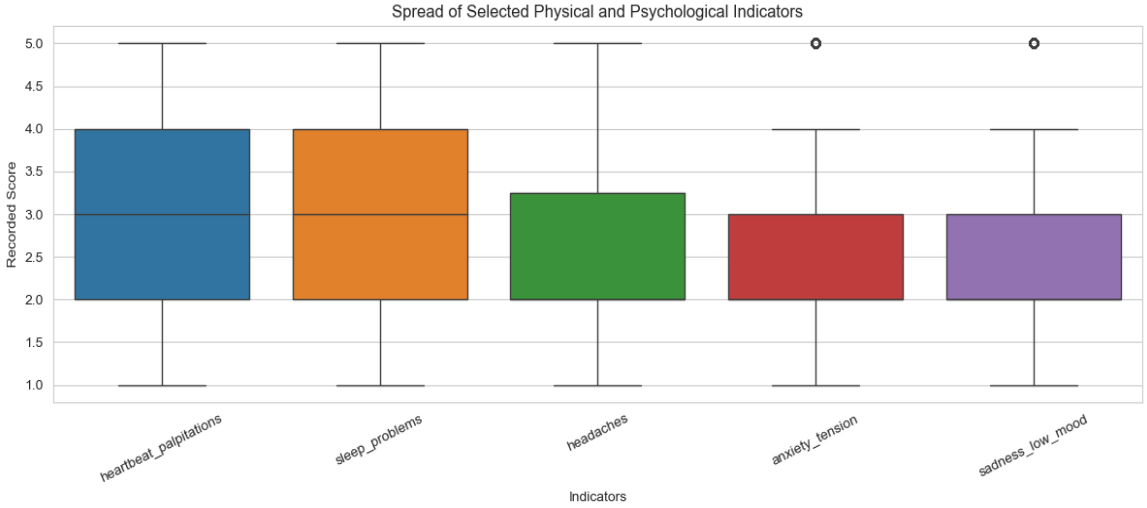


Figure 5. Spread of selected physical and psychological indicators.

The selected variables are recorded on small integer scales. Their boxplots therefore show repeated score levels rather than continuous measurements. This chart is useful for comparing central range and variation, but it should not be interpreted as a medical severity scale.

The chart also helps connect the physical and psychological variable groups to the first three project objectives.

3.5 RELATIONSHIP ANALYSIS: ANXIETY AND SLEEP PROBLEMS

The scatterplot compares anxiety_tension with sleep_problems. Each point represents a record, and colour indicates the coded stress_type class. The plot contains overlapping points because both variables take a small number of integer values.

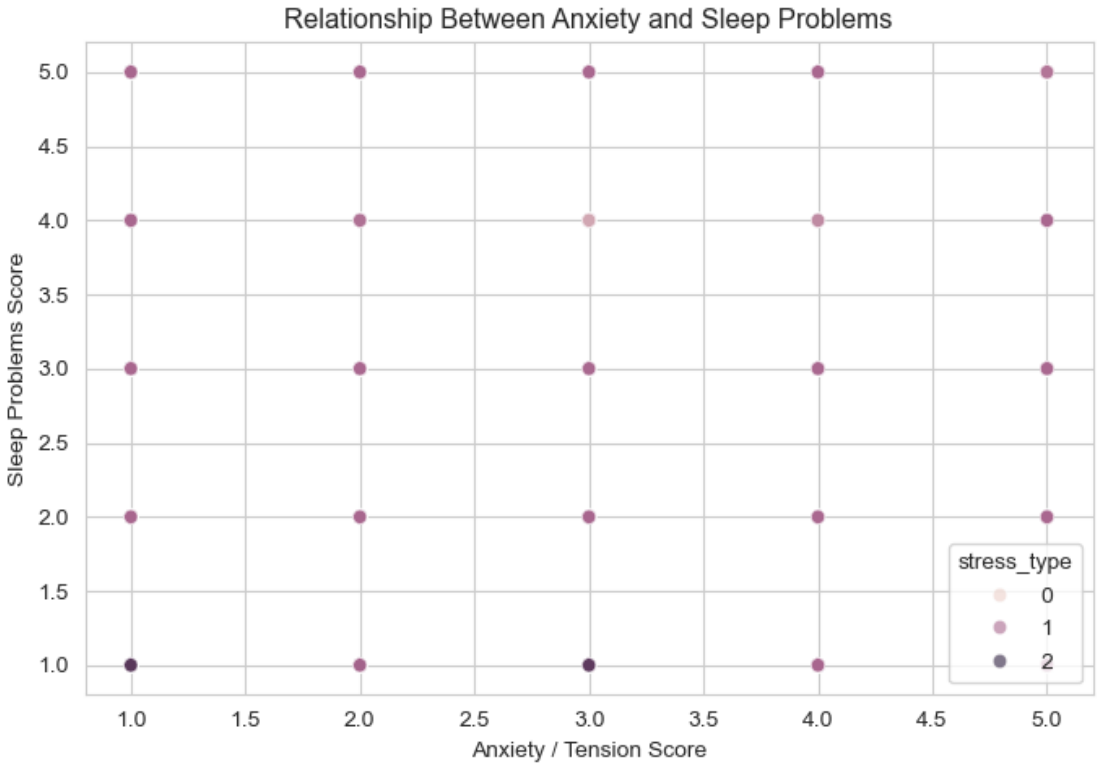


Figure 6. Relationship between anxiety/tension and sleep-problem scores.

The points occur at integer score combinations, so several records overlap at the same coordinates. The visualisation can show whether higher values of one variable tend to appear with higher values of the other, but it does not establish a causal relationship.

The relationship is also affected by the large number of records in class 1. Colour overlap should therefore be considered alongside the class distribution and correlation analysis.

3.6 RELATIONSHIP ANALYSIS: ACADEMIC OVERLOAD AND STRESS TYPE

Academic overload is compared across the three stress_type categories using a violin plot. The violin plot shows the distribution shape and quartile positions at different score levels. The comparison is descriptive and does not establish causation.

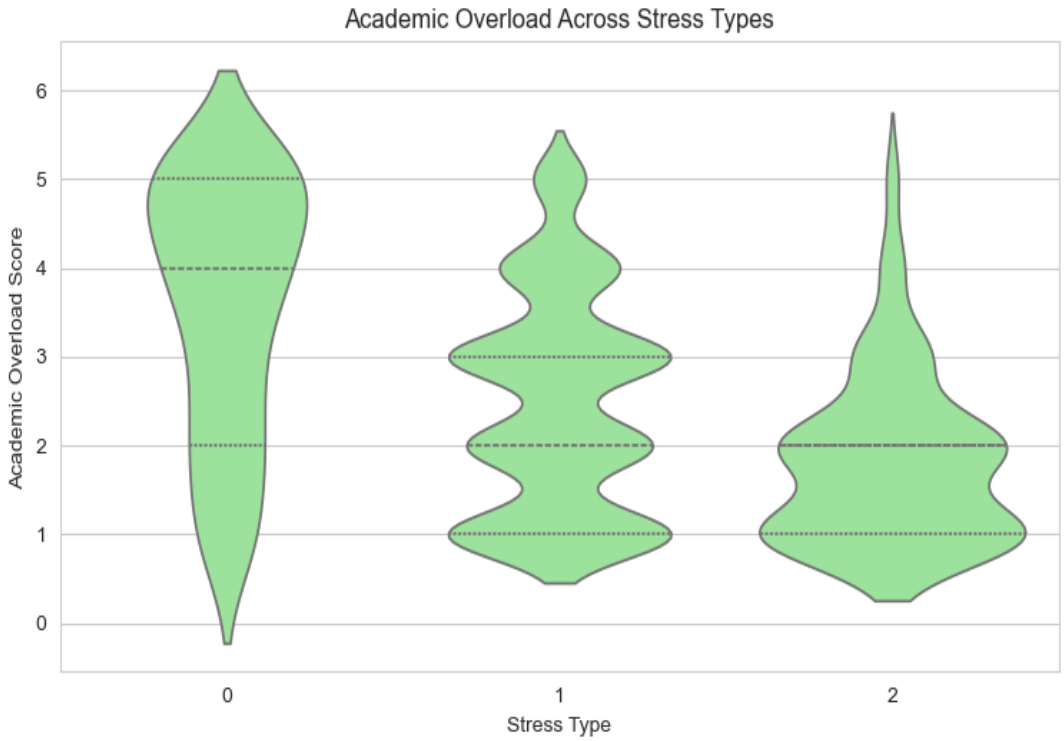


Figure 7. Academic overload across coded stress types.

These charts support the fourth project objective, which examines relationships between variables. They are descriptive comparisons only. They do not show that academic overload causes a particular stress category, and the coded target classes are not assigned unverified descriptive names.

3.7 CORRELATION ANALYSIS

A correlation matrix was calculated for seven selected numeric columns: anxiety_tension, sleep_problems, restlessness, sadness_low_mood, academic_overload, relationship_stress and stress_type. Correlation values range from -1 to +1 and describe pairwise linear association.

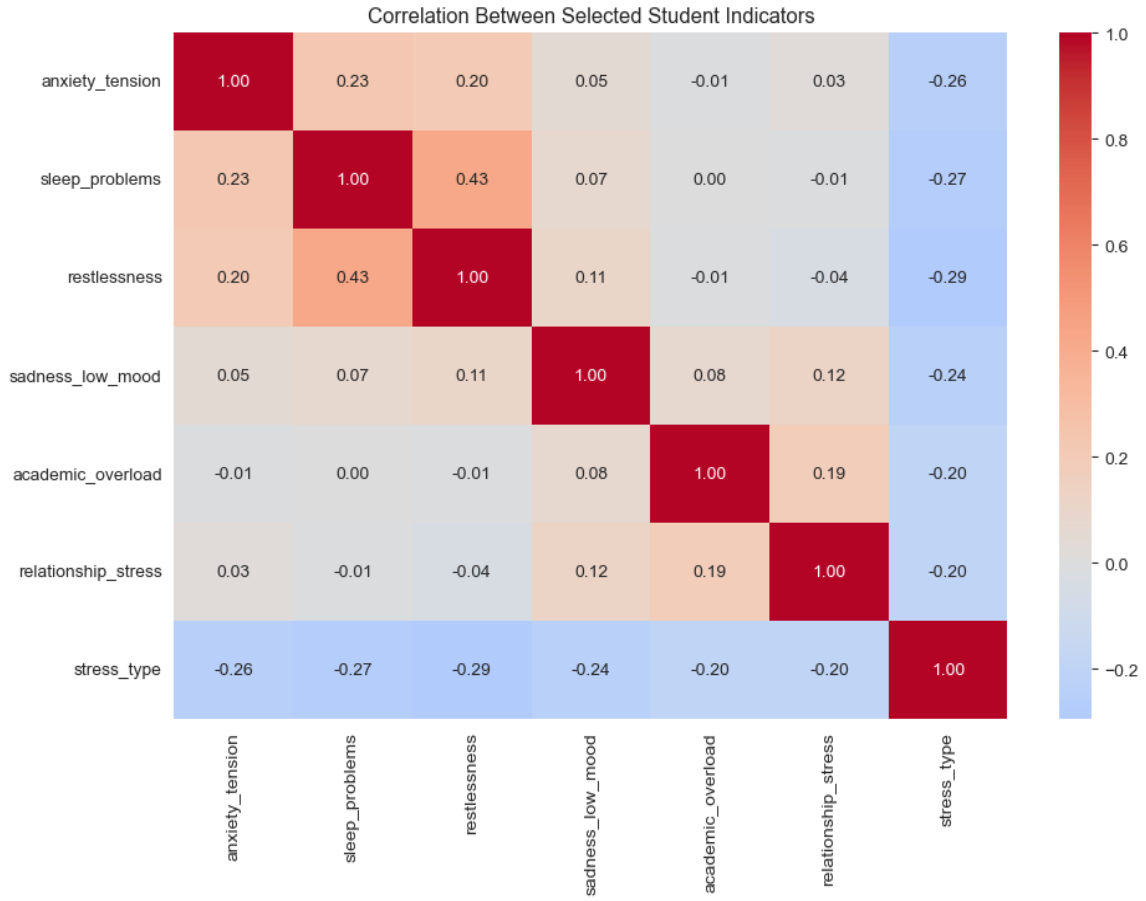


Figure 8. Correlation heatmap of selected numerical variables.

In the selected-variable heatmap, sleep_problems and restlessness have a positive correlation of approximately 0.43. Anxiety_tension and sleep_problems have a correlation of approximately 0.23. These associations do not establish cause and effect.

The heatmap includes stress_type as a numeric code for exploratory display. Since it is a categorical target represented by numbers, its Pearson correlation values should be treated cautiously and not as proof of a linear relationship with a continuous outcome.

4. ANALYSIS ON DATASET: PREPROCESSING

The target column is separated from the predictor columns. The project defines y as stress_type. The input matrix X contains all columns except stress_type and stress_experience. The latter is excluded because it may be too closely connected with the target and could provide information that would not be fair to use for prediction.

The resulting input matrix contains 24 features. The dataset columns are numeric in the supplied CSV, so a separate categorical encoding step was not required for this version of the project. The selected features include demographic, physical, psychological, academic and social indicators.

The data is split into training and testing sets with test_size=0.20 and random_state=42. The notebook does not specify stratify=y. This produces 1,600 training records and 400 test records. The training set is used to fit the models, while the test set is used to calculate accuracy on the held-out portion of this dataset.

StandardScaler is fitted only on the training data and then applied to the test data. This avoids using test-set information when calculating the scaling parameters.

4.1 CLASSIFICATION MODELS: OVERVIEW

Four classification algorithms were selected for comparison. The algorithms were chosen because they are standard introductory machine-learning methods and are covered in common Predictive Analytics learning materials.

Model	Basic working idea	Input handling
SVM	Finds a decision boundary that separates classes.	Scaled features
Naive Bayes	Uses class probabilities and feature likelihoods.	Scaled features in this experiment
KNN	Predicts from the classes of nearby training records.	Scaled features
Random Forest	Combines predictions from multiple decision trees.	Original numeric features

The models are trained using Scikit-learn. The implementation uses fit() for training and predict() for class predictions. The supplied notebook calculates confusion matrices and accuracy scores for each classifier. It does not calculate probability-based or averaged precision/recall metrics in the current version.

The comparison is based on the same original 2,000-record dataset and the same non-stratified train-test split. Because exact duplicate rows are retained, the split may place identical records in both subsets, which is a potential source of data leakage and inflated accuracy.

4.2 MODEL 1: SUPPORT VECTOR MACHINE (SVM)

The full SVM classifier uses an RBF kernel (kernel='rbf'), probability=True and random_state=42. It is trained using the scaled training features. A separate linear SVM with C=100 is trained on only anxiety_tension and sleep_problems for the two-dimensional decision-boundary illustration.

The full SVM model uses all 24 selected input features. A separate two-feature SVM is trained only for the decision-region visualisation. This graph uses anxiety_tension and sleep_problems so that the separation can be shown on a two-dimensional plot. The two-feature visualisation is not the model used for the final accuracy comparison.

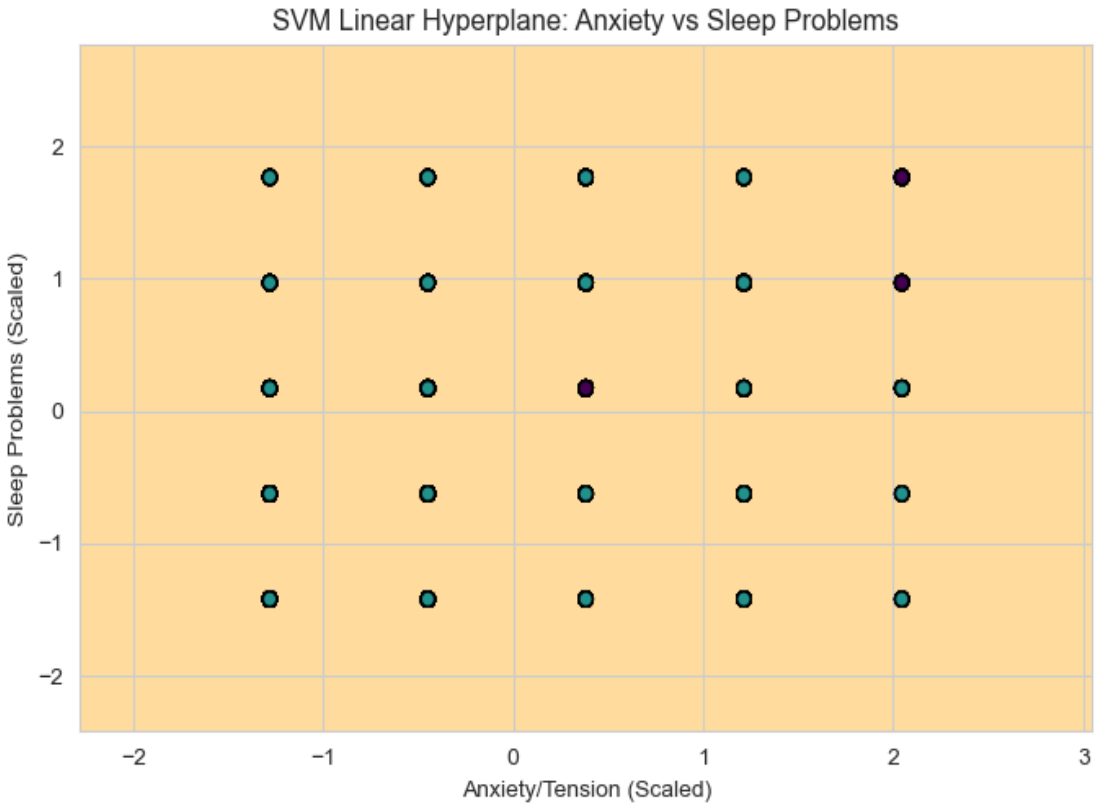


Figure 9. SVM decision boundary using anxiety/tension and sleep-problem scores.

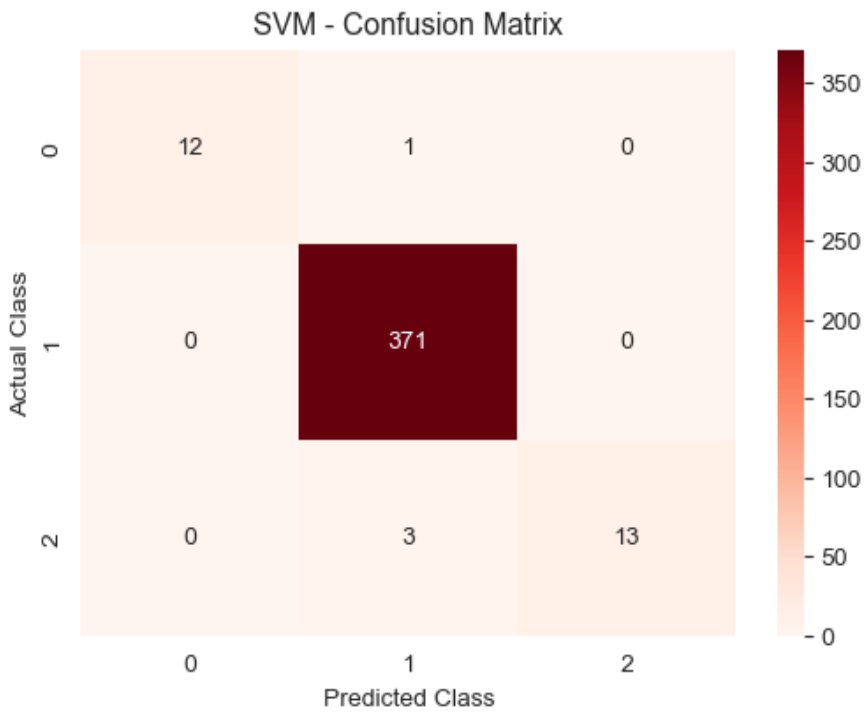
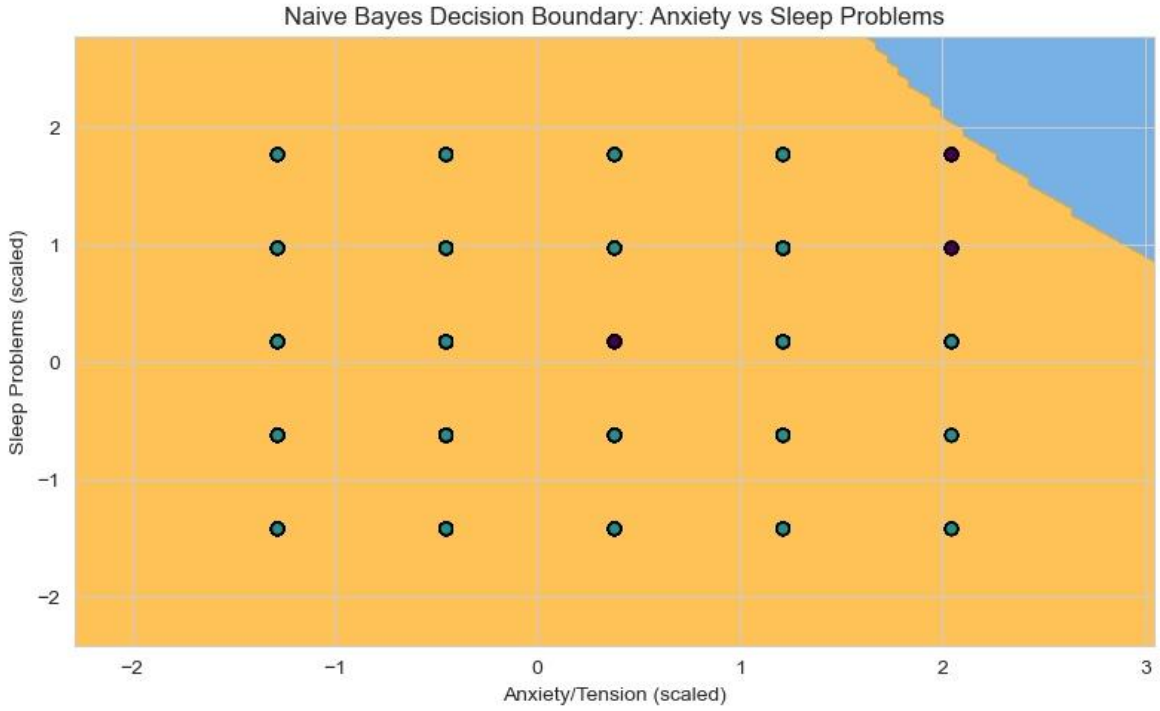


Figure 10. SVM confusion matrix on the 400-record test set.

The full SVM achieved 99.00% test accuracy (396 correct predictions out of 400). Its confusion matrix is [[12, 1, 0], [0, 371, 0], [0, 3, 13]], with rows representing actual classes 0, 1 and 2 and columns representing predicted classes 0, 1 and 2.

4.3 MODEL 2: NAIVE BAYES

Gaussian Naive Bayes is a probabilistic classifier. It uses Bayes' theorem and assumes that the input features are conditionally independent within each class. This assumption is simple and may not fully represent relationships between survey variables, but the method is useful as a basic comparison model.



The Gaussian Naive Bayes model is trained on the scaled training data and predicts the stress_type category for the test records. Its confusion matrix shows the class-wise prediction counts.

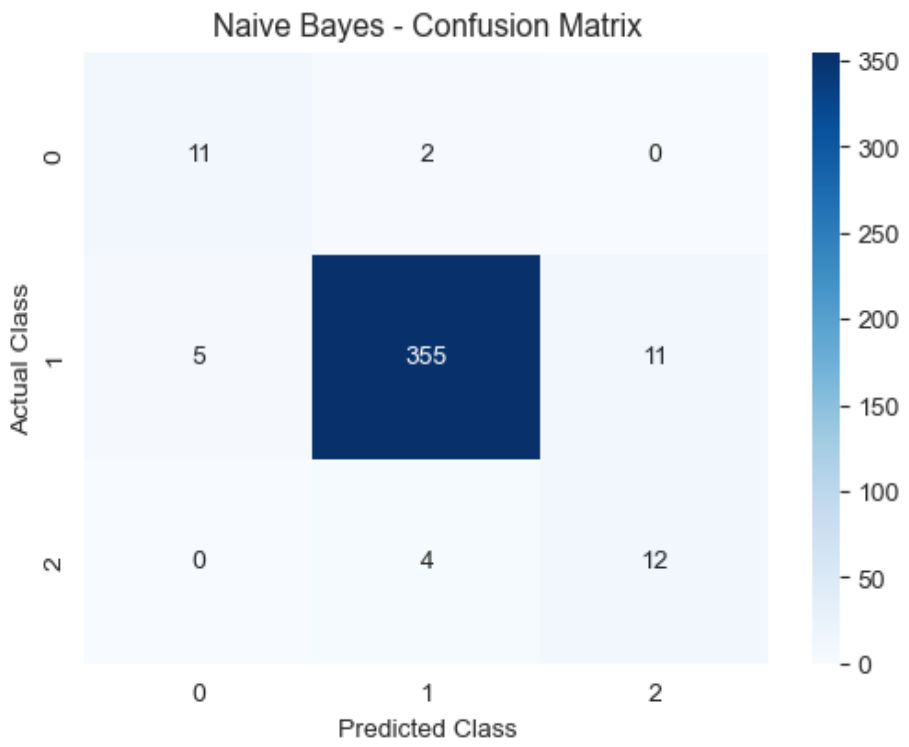


Figure 11. Naive Bayes confusion matrix.

Naive Bayes achieved 94.50% test accuracy (378 correct predictions out of 400). Its confusion matrix is [[11, 2, 0], [5, 355, 11], [0, 4, 12]].

4.4 MODEL 3: K-NEAREST NEIGHBORS (KNN)

K-Nearest Neighbors is a simple classification method. For a new record, it finds the nearest training records according to the selected distance measure and predicts the class that is most common among the neighbours. In this project, n_neighbors is set to 5.

Scaling is important for KNN because distance calculations can be affected when features have different numeric ranges. The scaled training and test features are therefore used for the main KNN model.

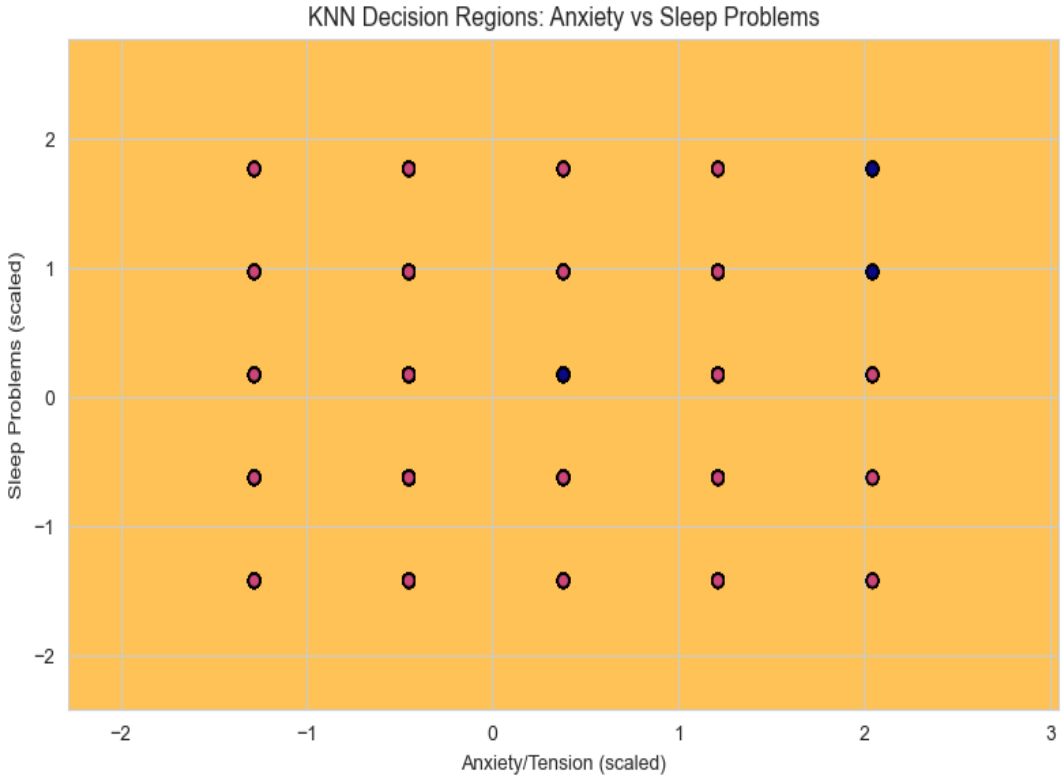


Figure 12. KNN decision regions using two selected variables.

The full KNN model achieved 94.75% test accuracy (379 correct predictions out of 400). The decision-region plot is a two-feature illustration using anxiety_tension and sleep_problems, while the reported accuracy comes from all 24 input features.

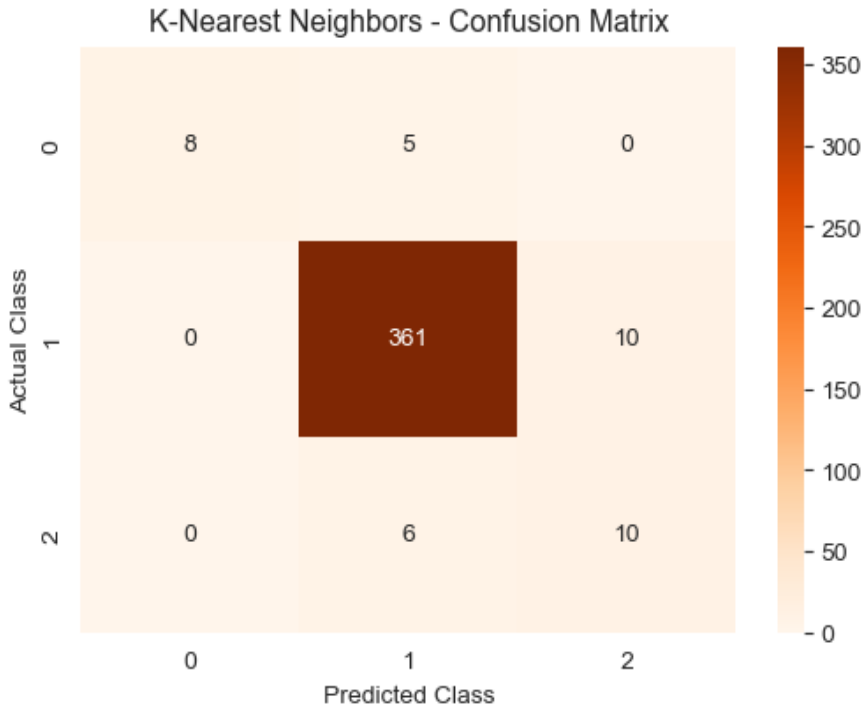


Figure 13. KNN confusion matrix.

4.5 MODEL 4: RANDOM FOREST

Random Forest is an ensemble classification method. It builds multiple decision trees and combines their predictions. In this project, the classifier uses n_estimators=100 and random_state=42.

Random Forest is trained on the original numeric feature values; the notebook does not scale inputs for this tree-based model. The first tree from the forest is displayed with max_depth=3 so that the diagram remains readable.

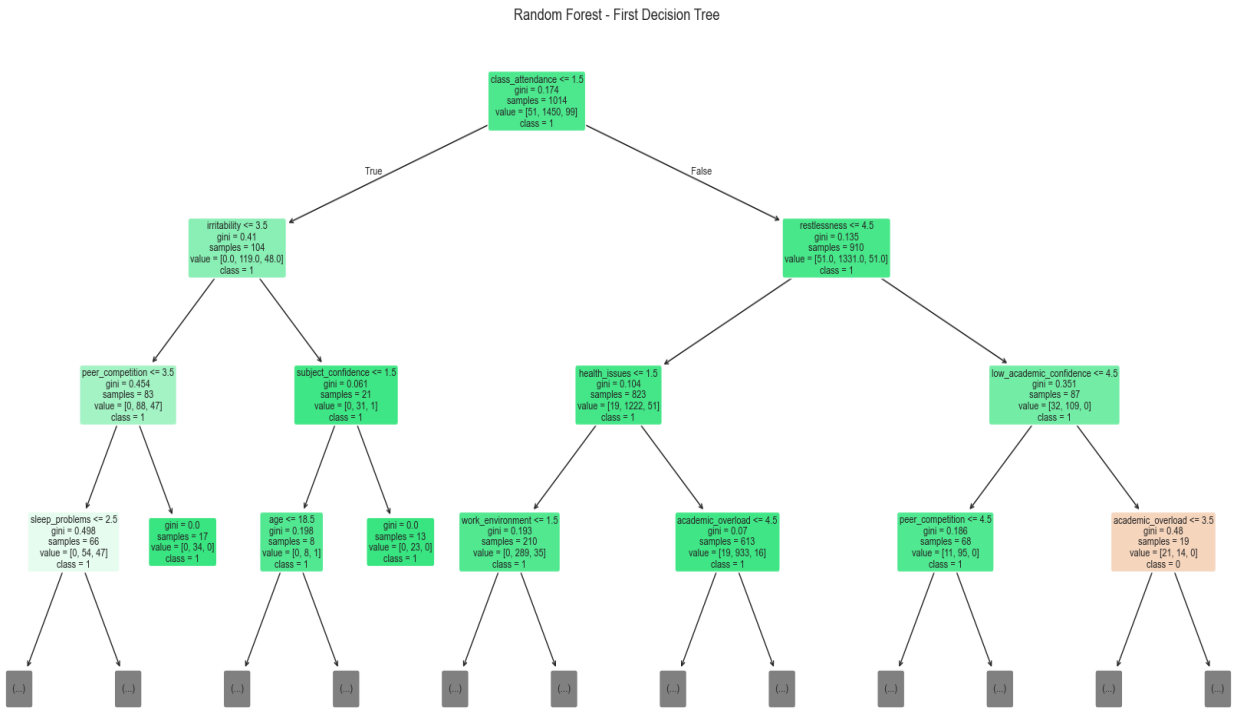


Figure 14. First decision tree from the Random Forest model (displayed to depth 3).

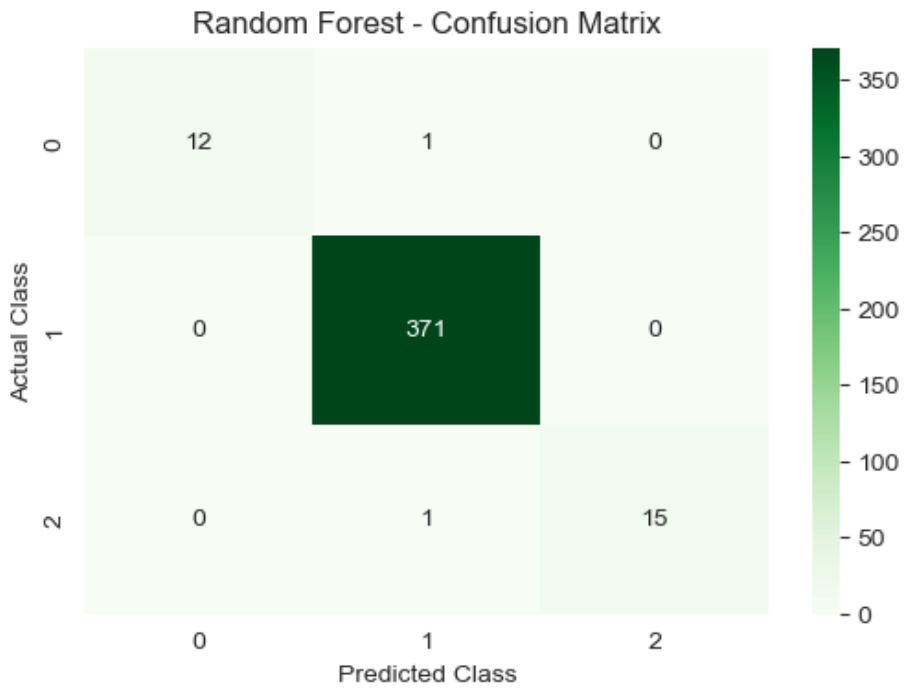


Figure 15. Random Forest confusion matrix.

Random Forest achieved 99.50% test accuracy (398 correct predictions out of 400). Its confusion matrix is [[12, 1, 0], [0, 371, 0], [0, 1, 15]]. The supplied notebook does not calculate or plot feature importance, so no feature-importance result is reported.

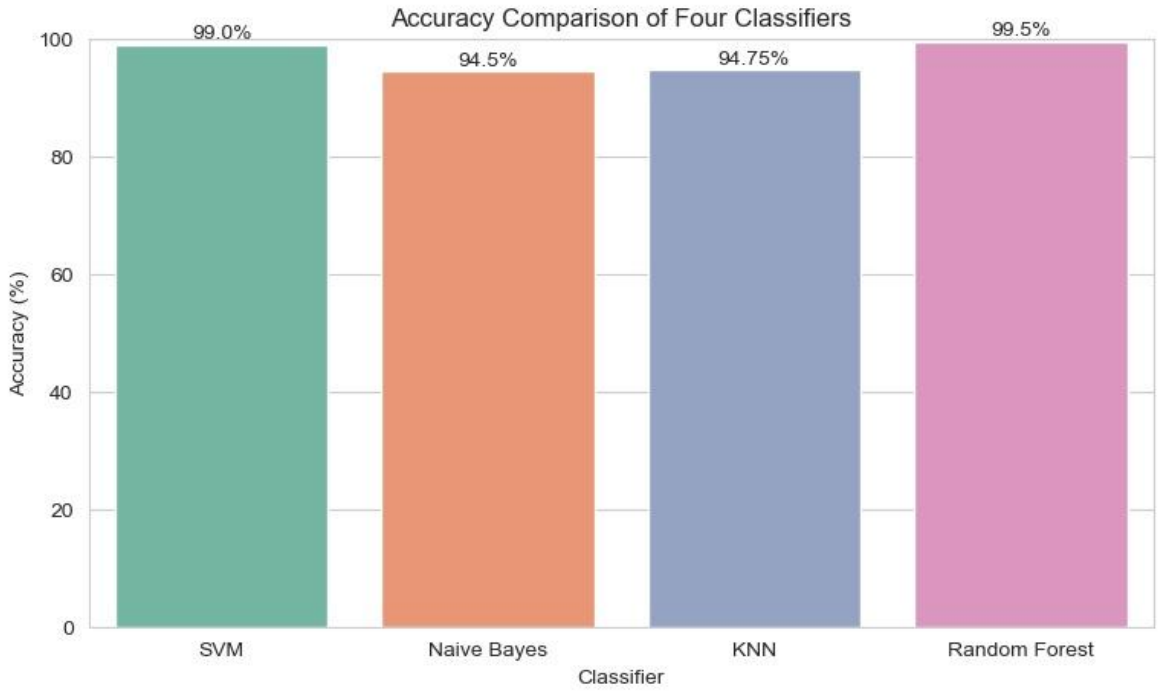


Figure 16. Accuracy comparison of the four classifiers.

4.7 IDENTIFICATION OF THE BEST-FIT MODEL

For this project, the best-fit classifier is selected using test accuracy. The highest accuracy is 99.50%, achieved by Random Forest. Therefore, Random Forest is identified as the best-fit model for this particular notebook run and train-test split.

Accuracy is the number of correct predictions divided by the total number of test predictions. However, the target data is highly imbalanced, so accuracy should not be viewed alone when judging practical usefulness. The confusion matrices show the prediction counts for all three classes.

The reported result is specific to the original dataset, selected features, random_state=42 and the particular non-stratified split. Since 1,191 duplicate rows are retained, identical records may occur in both training and test subsets. This can inflate test accuracy, so the 99.50% result should not be interpreted as validated performance on independent student data.

Random Forest is selected because it has the highest recorded test accuracy (99.50%), followed by SVM (99.00%), KNN (94.75%) and Naive Bayes (94.50%). A future rerun should remove or otherwise handle duplicate records before splitting and should use a stratified split or group-aware validation as appropriate.

```
# As required for this project, the best-fit model is selected using accuracy only.
best_accuracy = results['Accuracy'].max()
best_models = results.loc[results['Accuracy'] == best_accuracy, 'Classifier'].tolist()
print('Highest accuracy:', round(best_accuracy * 100, 2), '%')
print('Best-fit model(s) based on accuracy:', ', '.join(best_models))
```

```
Highest accuracy: 99.5 %
Best-fit model(s) based on accuracy: Random Forest
```

4.8 WHY REGRESSION IS NOT APPLIED

Regression is used when the target variable is a continuous numerical value, such as a measured quantity. In this dataset, the target variable stress_type contains three category codes: 0, 1 and 2. These values represent classes rather than a continuous measurement.

For this reason, ordinary Linear Regression is not applied to stress_type. Treating the class codes as continuous numbers would introduce an incorrect assumption about the distance between categories. Regression metrics such as MAE, MSE, RMSE and R-squared are therefore not reported for this classification target.

A future regression extension would require a genuine numerical outcome, such as a consistently collected energy score. The supplied dataset does not include such a continuous target in the current project file. The report therefore explains the difference between classification and regression rather than forcing an unsuitable regression model.

This decision keeps the project aligned with the actual structure of the data.

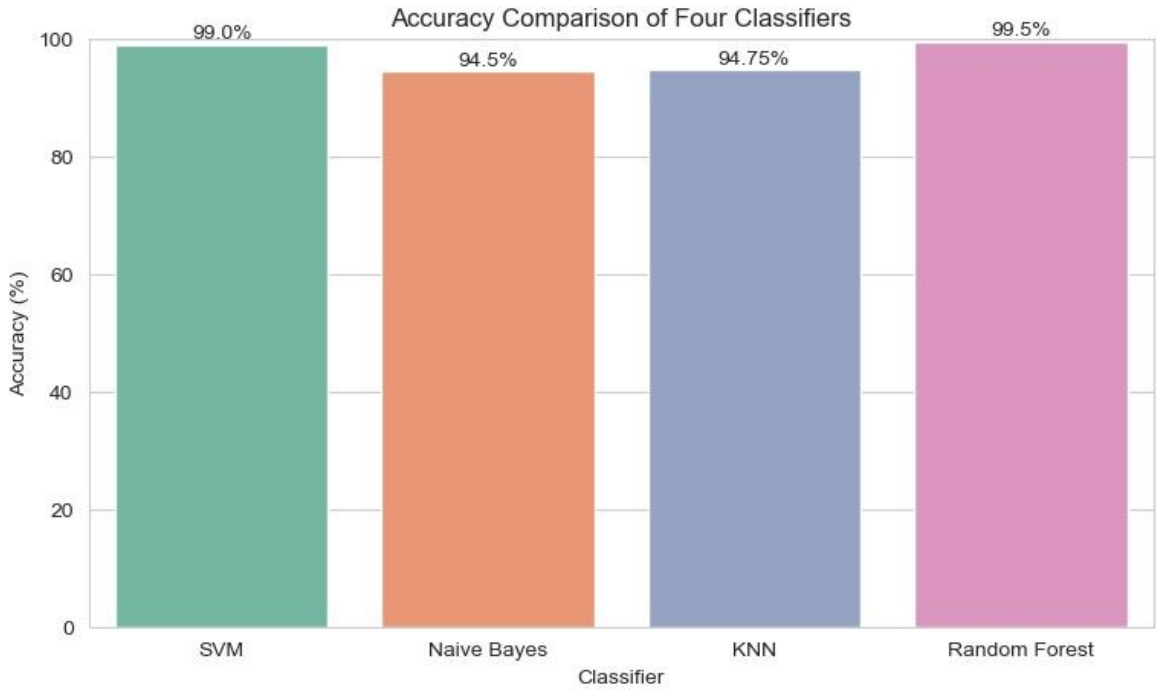
5. CONCLUSION

MentalBridge applied Predictive Analytics methods to a student stress-indicator survey dataset. The work included data loading, inspection, missing-value and duplicate checks, exploratory visualisation, feature selection, preprocessing and comparison of four classification algorithms.

The notebook loaded 2,000 records and 26 columns. It found zero missing values and detected 1,191 exact duplicate rows, but did not remove them. The target variable stress_type had three coded classes: 0 (70 records), 1 (1,828 records) and 2 (102 records). The feature stress_experience was excluded from the model inputs, leaving 24 predictors.

Random Forest achieved the highest test accuracy at 99.50%, followed by SVM at 99.00%, KNN at 94.75% and Naive Bayes at 94.50%. Random Forest was therefore selected as the best-fit model based on accuracy for this experiment. The test set contained 400 records.

The results demonstrate the practical steps involved in a predictive analytics workflow. However, the detected duplicates were retained and the split was not stratified. These choices may inflate the reported accuracy, particularly because identical records can appear in both subsets. The results should be treated as preliminary academic findings, not as validated generalisation performance or a medical assessment.



6. FUTURE SCOPE

The project can be improved in several ways while keeping the same basic approach. First, the dataset documentation can be studied in more detail to confirm the meaning of all coded values and the way records were collected. This would make the interpretation of gender and target categories more precise.

Second, the models can be evaluated using repeated stratified cross-validation instead of relying on one train-test split. This would show whether the model results remain similar across different partitions. Any such extension should continue to avoid leakage between training and testing data.

Third, class imbalance can be studied using class-wise precision, recall, F1-score and confusion matrices. If resampling or class weighting is explored, it should be applied only to the training data and its effect should be compared transparently.

Fourth, additional genuine data collected with consent and appropriate privacy safeguards could support a separate analysis of changes over time. The current dataset does not contain menstrual-cycle dates or a continuous daily energy target, so cycle forecasting and energy regression are not included in this report.

Finally, the work could be presented through a simple notebook or educational dashboard, provided that any future interface clearly explains that predictions are experimental and not clinical advice.

7. REFERENCES

28. Stress Indicators Dataset for Mental Health Classification. Mendeley Data. Available at: https://data.mendeley.com/datasets/2gsjv8m7ch/2 (accessed 31 October 2026).
29. McKinney, W. (2022). Python for Data Analysis: Data Wrangling with pandas, NumPy, and Jupyter. 3rd edition. O'Reilly Media.
30. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
31. Scikit-learn documentation. Supervised learning and model evaluation. https://scikit-learn.org/stable/
32. Pandas documentation. https://pandas.pydata.org/docs/
33. NumPy documentation. https://numpy.org/doc/
34. Matplotlib documentation. https://matplotlib.org/stable/
35. Seaborn documentation. https://seaborn.pydata.org/

The library documentation was used as a reference for the Python functions and machine-learning implementations used in the notebook. Dataset details should be checked against the Mendeley Data landing page at the time of final submission.

IMPORTANT FUNCTIONS USED

Function / method	Purpose in this project
pd.read_csv()	Reads the CSV dataset into a DataFrame.
df.info()	Displays column names, data types and non-null counts.
df.describe()	Produces descriptive statistics for numeric columns.
df.duplicated()	Identifies exact duplicate rows.
df.duplicated().sum()	Counts exact duplicate rows; duplicates are not removed in this notebook.
train_test_split()	Divides records into training and testing subsets.
StandardScaler()	Standardises selected numeric input features.
.fit()	Trains a machine-learning model.
.predict()	Generates predicted class labels.
accuracy_score()	Calculates the proportion of correct predictions.
confusion_matrix()	Summarises actual and predicted class counts.
sns.heatmap()	Displays values using a colour-coded matrix.

These functions are included to make the implementation understandable and to connect the written report with the practical notebook.

LIMITATIONS AND INTERPRETATION

The notebook uses all 2,000 rows, including 1,191 exact duplicate rows detected during data inspection. Class 1 is much more common than classes 0 and 2. This imbalance affects interpretation of overall accuracy; confusion matrices should also be consulted.

The results come from a single non-stratified 80:20 split with random_state=42 (1,600 training records and 400 test records). Duplicate records were not removed before splitting, so train-test overlap of identical rows is possible. The results are not a guarantee of performance on new institutions, time periods or individuals outside the dataset.

The numeric values for gender and stress_type are treated as coded categories. The report does not assign labels that are not confirmed by the supplied data documentation. The feature stress_experience is excluded from the model inputs. The duplicate count is reported accurately, but duplicates remain in the notebook's modelling data.

MentalBridge is an academic predictive analytics exercise. It is not a medical tool and should not be used to diagnose, treat or make decisions about a person's health.

CODE

```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns


from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.svm import SVC

from sklearn.naive_bayes import GaussianNB

from sklearn.neighbors import KNeighborsClassifier

from sklearn.ensemble import RandomForestClassifier
```

```
from sklearn.inspection import DecisionBoundaryDisplay
from sklearn.tree import plot_tree
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    confusion_matrix, classification_report
)
sns.set_style("whitegrid")
plt.rcParams["figure.figsize"] = (8, 5)



df = pd.read_csv("/Users/aryasrivastava23/Documents/Stress Indicators Dataset for Mental Health Classification.csv")
print("Shape of dataset:", df.shape)
df.head(10)

#DEScriptive
df.tail(5)
df.info()
print("\nStatistical summary:")
display(df.describe())
print("\nMissing values in each column:")
print(df.isnull().sum())
print("Number of duplicate rows:", df.duplicated().sum())
print("\\nOriginal target class counts:")
print(df["stress_type"].value_counts().sort_index())
```

```
#possible outliers
print("\\nSkewness of numeric columns:")
print(df.select_dtypes(include=np.number).skew().sort_values(ascending=False).to_frame("Skewness"))

#AGE DISTRIBUTION AND GENDER CATEGORY COUNT
fig, axes = plt.subplots(1, 2, figsize=(12, 4))
sns.histplot(data=df, x='age', bins=12, kde=True, color='cornflowerblue', ax=axes[0])
axes[0].set_title('Age Distribution')
sns.countplot(data=df, x='gender', hue='gender', palette='pastel', legend=False, ax=axes[1])
axes[1].set_title('Gender Category Count')
plt.tight_layout()
plt.show()

#Spread of Selected Physical and Psychological Indicators
selected_indicators = [
    "heartbeat_palpitations", "sleep_problems",
    "headaches", "anxiety_tension", "sadness_low_mood"
]
plt.figure(figsize=(12, 5))
sns.boxplot(data=df[selected_indicators])
plt.title("Spread of Selected Physical and Psychological Indicators")
plt.xlabel("Indicators")
plt.ylabel("Recorded Score")
plt.xticks(rotation=25)
plt.tight_layout()
```

```
plt.show()


#Spread of Selected Physical and Psychological Indicators
selected_indicators = [
    "heartbeat_palpitations", "sleep_problems",
    "headaches", "anxiety_tension", "sadness_low_mood"
]
plt.figure(figsize=(12, 5))
sns.boxplot(data=df[selected_indicators])
plt.title("Spread of Selected Physical and Psychological Indicators")
plt.xlabel("Indicators")
plt.ylabel("Recorded Score")
plt.xticks(rotation=25)
plt.tight_layout()
plt.show()


#Academic Overload Across Stress Types
plt.figure(figsize=(9, 5))
sns.violinplot(
    data=df,
    x="stress_type",
    y="academic_overload",
    inner="quartile",
    color="lightgreen"
)
plt.title("Academic Overload Across Stress Types")
```

```
plt.xlabel("Stress Type")
plt.ylabel("Academic Overload Score")
plt.show()
selected_features = ['anxiety_tension', 'sleep_problems', 'restlessness', 'academic_overload']
df[selected_features].hist(figsize=(10, 7), bins=8, color='skyblue', edgecolor='black')
plt.suptitle('Distribution of Selected Stress Indicators')
plt.tight_layout()
plt.show()

#CORRELATION HEATMAP
relationship_columns = [
    "anxiety_tension", "sleep_problems", "restlessness",
    "sadness_low_mood", "academic_overload",
    "relationship_stress", "stress_type"
]
relationship_corr = df[relationship_columns].corr()
plt.figure(figsize=(10, 7))
sns.heatmap(relationship_corr, annot=True, cmap="coolwarm", center=0, fmt=".2f")
plt.title("Correlation Between Selected Student Indicators")
plt.tight_layout()
plt.show()
```

```
#Preprocessing for Classification

#Inferential

target = "stress_type"

X = df.drop(columns=[target, "stress_experience"])

y = df[target]

# Fill missing numeric values

X = X.fillna(X.median(numeric_only=True))

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.20, random_state=42)

s = StandardScaler()

X_train_scaled = s.fit_transform(X_train)

X_test_scaled = s.transform(X_test)


#CLASSIFIER IMPLEMENTATION

# 1. Support Vector Machine

svm_model = SVC(kernel="rbf", probability=True, random_state=42)

svm_model.fit(X_train_scaled, y_train)

svm_pred = svm_model.predict(X_test_scaled)

cm = confusion_matrix(y_test,svm_pred)

svm_acc = accuracy_score(y_test,svm_pred)

print("\nConfusion confusion_matrix: ")

print(cm)

print("\nAccuracy:", svm_acc)


# Select two variables only for the 2-D visualisation

visual_features = ["anxiety_tension", "sleep_problems"]
```

```
X_train_two = X_train[visual_features]
X_test_two = X_test[visual_features]
# Scale the two selected variables
visual_scaler = StandardScaler()
X_train_two_scaled = visual_scaler.fit_transform(X_train_two)

# Train a simple SVM model for this graph
# SVM Linear Hyperplane

svm_visual = SVC(kernel='linear', C=100)
svm_visual.fit(X_train_two_scaled, y_train)
plt.figure(figsize=(9, 6))
DecisionBoundaryDisplay.from_estimator(
    svm_visual,
    X_train_two_scaled,
    response_method='predict',
    cmap='coolwarm',
    alpha=0.4
)
plt.scatter(
    X_train_two_scaled[:, 0],
    X_train_two_scaled[:, 1],
    c=y_train,
    cmap='viridis',
    edgecolors='black'
```

```
)
plt.title("SVM Linear Hyperplane: Anxiety vs Sleep Problems")
plt.xlabel("Anxiety/Tension (Scaled)")
plt.ylabel("Sleep Problems (Scaled)")
plt.show()
plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, svm_pred), annot=True, fmt='d', cmap='Reds')
plt.title('SVM - Confusion Matrix')
plt.xlabel('Predicted Class')
plt.ylabel('Actual Class')
plt.show()

# 2. Naive Bayes
nb_model = GaussianNB()
nb_model.fit(X_train_scaled, y_train)
nb_pred = nb_model.predict(X_test_scaled)

cm_nb = confusion_matrix(y_test, nb_pred)
acc_nb = accuracy_score(y_test, nb_pred)
print("\nNaive Bayes Confusion Matrix:")
print(cm_nb)
print("\nNaive Bayes Accuracy:", acc_nb)



# Naive Bayes Decision Boundary
```

```
nb_visual = GaussianNB()
nb_visual.fit(X_train_two_scaled, y_train)
plt.figure(figsize=(9, 6))
DecisionBoundaryDisplay.from_estimator(
    nb_visual,
    X_train_two_scaled,
    response_method="predict",
    cmap="Pastel2",
    alpha=0.7
)
plt.scatter(
    X_train_two_scaled[:, 0],
    X_train_two_scaled[:, 1],
    c=y_train,
    cmap="viridis",
    edgecolors="black",
    s=35
)
plt.title("Naive Bayes Decision Boundary: Anxiety vs Sleep Problems")
plt.xlabel("Anxiety/Tension (scaled)")
plt.ylabel("Sleep Problems (scaled)")
plt.tight_layout()
plt.show()
plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, nb_pred), annot=True, fmt='d', cmap='Blues')
plt.title('Naive Bayes - Confusion Matrix')
```

```
plt.xlabel('Predicted Class')
plt.ylabel('Actual Class')
plt.show()

# 3. K-Nearest Neighbors
knn_model = KNeighborsClassifier(n_neighbors=5, metric="euclidean")
knn_model.fit(X_train_scaled, y_train)
knn_pred = knn_model.predict(X_test_scaled)

cm_knn = confusion_matrix(y_test, knn_pred)
acc_knn = accuracy_score(y_test, knn_pred)
print("\nK-Nearest Neighbors Confusion Matrix:")
print(cm_knn)
print("\nK-Nearest Neighbors Accuracy:", acc_knn)

# Train a simple KNN model on the same two variables for visualisation
knn_visual = KNeighborsClassifier(n_neighbors=5, metric="euclidean")
knn_visual.fit(X_train_two_scaled, y_train)

plt.figure(figsize=(9, 6))
DecisionBoundaryDisplay.from_estimator(
    knn_visual,
    X_train_two_scaled,
    response_method="predict",
    cmap="Pastel2",
    alpha=0.7
```

```
)
plt.scatter(
    X_train_two_scaled[:, 0],
    X_train_two_scaled[:, 1],
    c=y_train,
    cmap="plasma",
    edgecolors="black",
    s=35
)
plt.title("KNN Decision Regions: Anxiety vs Sleep Problems")
plt.xlabel("Anxiety/Tension (scaled)")
plt.ylabel("Sleep Problems (scaled)")
plt.tight_layout()
plt.show()

plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, knn_pred), annot=True, fmt='d', cmap='Oranges')
plt.title('K-Nearest Neighbors - Confusion Matrix')
plt.xlabel('Predicted Class')
plt.ylabel('Actual Class')
plt.show()

# 4. Random Forest
rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
rf_model.fit(X_train, y_train)
rf_pred = rf_model.predict(X_test)
```

```
cm = confusion_matrix(y_test, rf_pred)
print("Confusion Matrix:",cm)
acc = accuracy_score(y_test, rf_pred)
print("Accuracy:", acc)

#RANDOM FOREST DIAGRAM
# Display the first tree from the trained Random Forest
plt.figure(figsize=(22, 10))
plot_tree(
    rf_model.estimators_[0],
    feature_names=X.columns,
    class_names=[str(c) for c in rf_model.classes_],
    filled=True,
    rounded=True,
    max_depth=3,
    fontsize=8
)
plt.title('Random Forest - First Decision Tree ')
plt.show()
plt.figure(figsize=(6, 4))
sns.heatmap(confusion_matrix(y_test, rf_pred), annot=True, fmt='d', cmap='Greens')
plt.title('Random Forest - Confusion Matrix')
plt.xlabel('Predicted Class')
plt.ylabel('Actual Class')
plt.show()
```

```
#IDENTIFY THE BEST MODEL BASED ON ACCURACY
results = pd.DataFrame({
    'Classifier': ['SVM', 'Naive Bayes', 'KNN', 'Random Forest'],
    'Accuracy': [svm_acc, acc_nb, acc_knn, acc]
})
results['Accuracy (%)'] = (results['Accuracy'] * 100).round(2)
display(results[['Classifier', 'Accuracy (%)']].sort_values('Accuracy (%)',
ascending=False))



plt.figure(figsize=(9, 5))
sns.barplot(data=results, x='Classifier', y='Accuracy (%)', hue='Classifier', palette='Set2',
legend=False)
plt.title('Accuracy Comparison of Four Classifiers')
plt.ylabel('Accuracy (%)')
plt.ylim(0, 100)
for i, value in enumerate(results['Accuracy (%)']):
    plt.text(i, value + 1, str(value) + '%', ha='center')
plt.show()



# As required for this project, the best-fit model is selected using accuracy only.
best_accuracy = results['Accuracy'].max()
best_models = results.loc[results['Accuracy'] == best_accuracy, 'Classifier'].tolist()
print('Highest accuracy:', round(best_accuracy * 100, 2), '%')
```

```
print('Best-fit model(s) based on accuracy:', ', '.join(best_models))
```

Linkedin:- https://lnkd.in/p/g9ysXyKm

Github:- https://github.com/aryasrivastava23/MentalBridge